

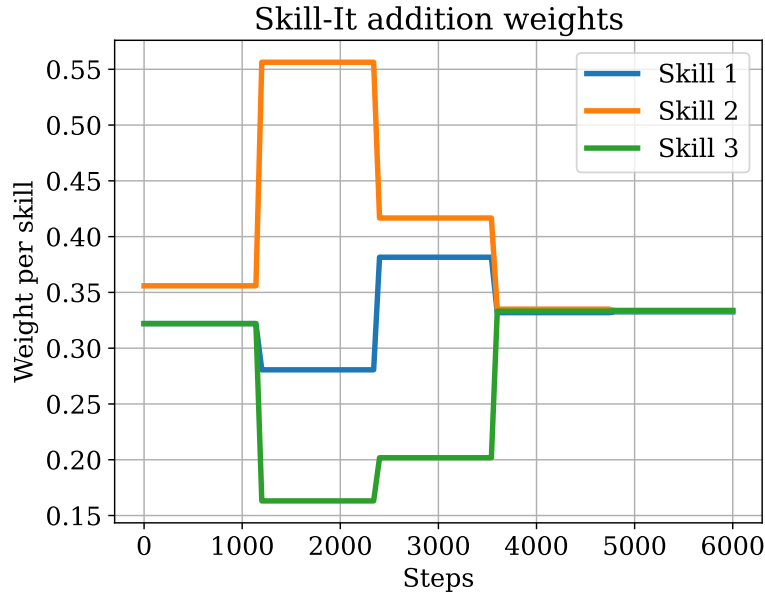


Figure 21: Weight per skill for addition pre-training experiment. SKILL-IT initially allocates more weight to skill 2, which has an edge to skill 1, while allocating little weight to skill 3 which is learned quickly. Eventually, SKILL-IT puts more weight on the harder skill 1 before converging to uniform sampling when all losses roughly approach 0.

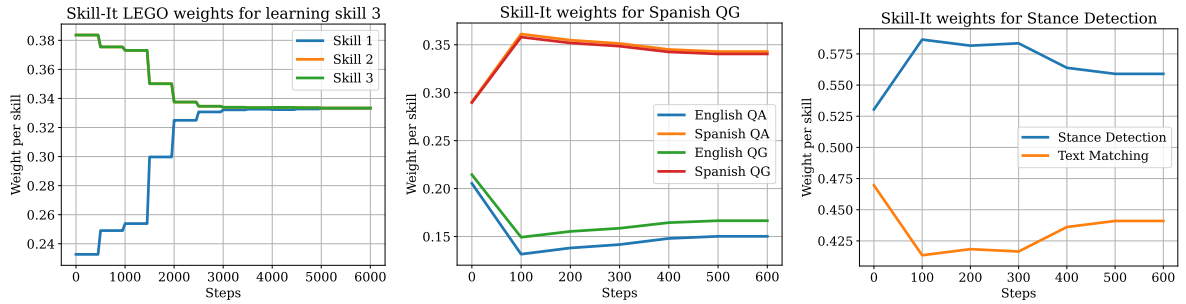


Figure 22: Weight per skill for fine-tuning experiments. Left: LEGO; Center: Spanish question generation; Right: stance detection.

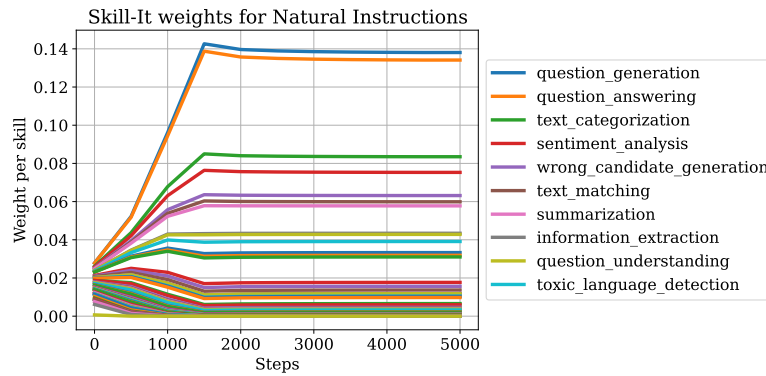


Figure 23: Weight per skill for Natural Instructions out-of-domain experiment. The legend shows the top 10 skills with the largest weight. While the relative order of weight magnitude does not change significantly across training, the incorporation of loss dramatically increases the range of the weights, showing the importance of an online algorithm.

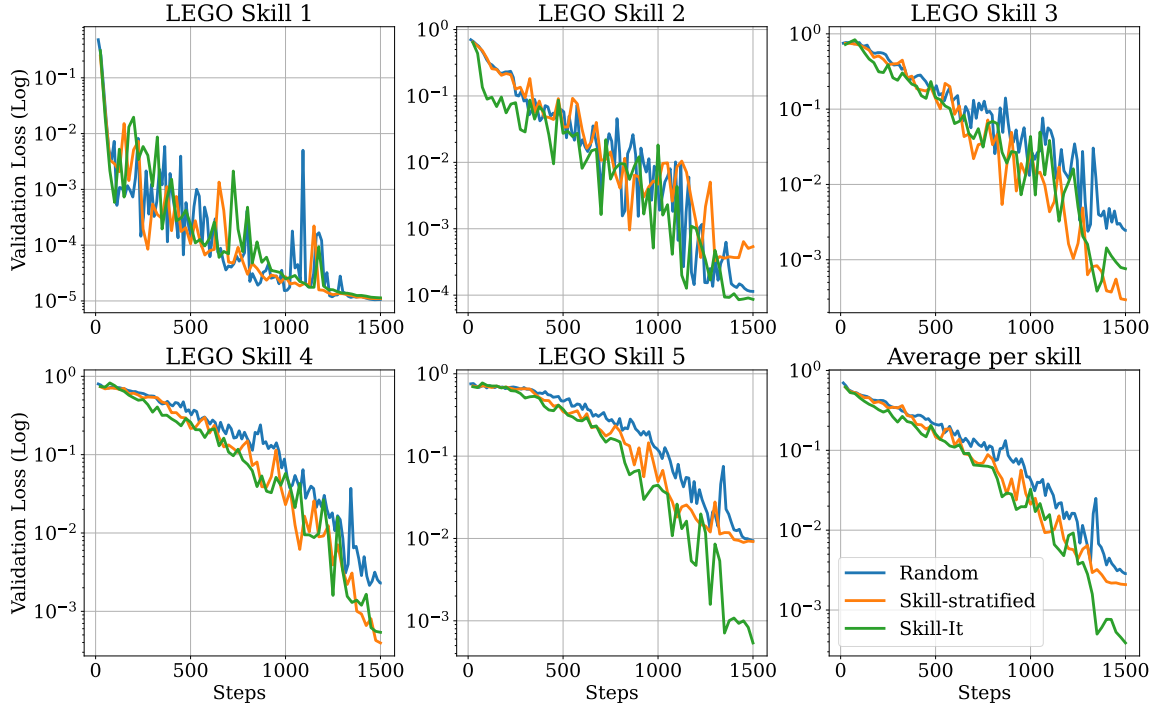


Figure 24: Performance of SKILL-IT for LEGO pre-training setting when skills graph is learned on a 125M parameter model and used for data selection with a 1.3B model. SKILL-IT on average still outperforms random and skill-stratified sampling, suggesting that findings on ordered skill sets can transfer from small models to large models.

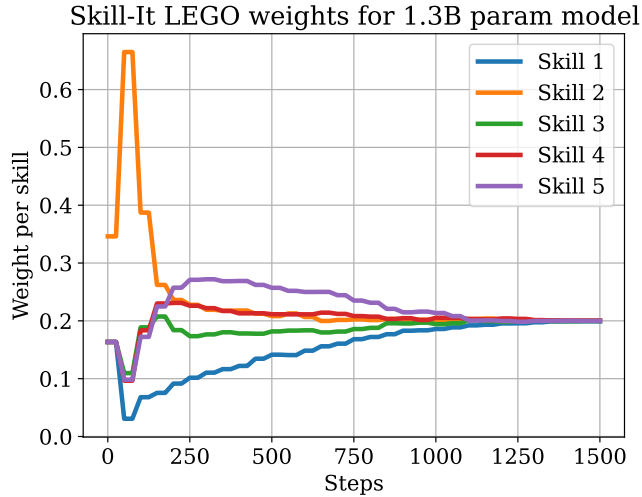


Figure 25: Weight per skill for LEGO pre-training experiment on 1.3B parameter model. The trajectories are similar to those of the 125M parameter model in Figure 20. SKILL-IT initially allocates more weight to skill 2, but eventually puts more weight on skills 4 and 5 before converging to uniform sampling when all losses converge to near 0.

Skill	Random	Skill-stratified	SKILL-IT
Answer Verification	2.005 $\pm$ 0.059	1.903 $\pm$ 0.069	1.890 $\pm$ 0.072
Code to Text	0.302 $\pm$ 0.032	0.204 $\pm$ 0.022	0.269 $\pm$ 0.032
Discourse Connective Identification	2.529 $\pm$ 0.046	2.372 $\pm$ 0.054	2.393 $\pm$ 0.056
Entity Generation	2.108 $\pm$ 0.328	1.788 $\pm$ 0.429	1.885 $\pm$ 0.461
Entity Relation Classification	1.130 $\pm$ 0.048	0.836 $\pm$ 0.006	0.841 $\pm$ 0.010
Information Extraction	2.032 $\pm$ 0.013	1.992 $\pm$ 0.006	1.933 $\pm$ 0.013
Irony Detection	2.802 $\pm$ 0.125	2.528 $\pm$ 0.146	2.585 $\pm$ 0.149
Preposition Prediction	1.095 $\pm$ 0.040	0.686 $\pm$ 0.041	0.774 $\pm$ 0.029
Punctuation Error Detection	2.633 $\pm$ 0.027	3.188 $\pm$ 0.055	2.726 $\pm$ 0.025
Question Answering	1.947 $\pm$ 0.003	2.119 $\pm$ 0.003	2.073 $\pm$ 0.001
Question Generation	2.214 $\pm$ 0.007	2.345 $\pm$ 0.008	2.263 $\pm$ 0.010
Question Understanding	1.928 $\pm$ 0.020	1.837 $\pm$ 0.031	1.700 $\pm$ 0.042
Sentence Expansion	2.054 $\pm$ 0.018	1.828 $\pm$ 0.060	1.853 $\pm$ 0.058
Sentiment Analysis	2.771 $\pm$ 0.009	2.818 $\pm$ 0.006	2.774 $\pm$ 0.007
Stance Detection	1.814 $\pm$ 0.151	1.500 $\pm$ 0.117	1.628 $\pm$ 0.149
Summarization	2.531 $\pm$ 0.009	2.472 $\pm$ 0.012	2.440 $\pm$ 0.013
Text Categorization	2.289 $\pm$ 0.016	2.341 $\pm$ 0.021	2.231 $\pm$ 0.022
Text Matching	1.967 $\pm$ 0.008	1.913 $\pm$ 0.005	1.872 $\pm$ 0.005
Text Simplification	1.861 $\pm$ 0.003	1.692 $\pm$ 0.023	1.698 $\pm$ 0.022
Text to Code	0.614 $\pm$ 0.030	0.518 $\pm$ 0.030	0.585 $\pm$ 0.022
Toxic Language Detection	2.853 $\pm$ 0.020	2.911 $\pm$ 0.019	2.862 $\pm$ 0.018
Word Semantics	1.999 $\pm$ 0.023	1.870 $\pm$ 0.039	1.902 $\pm$ 0.024
Wrong Candidate Generation	2.187 $\pm$ 0.028	2.192 $\pm$ 0.023	2.140 $\pm$ 0.020
Average	1.985 $\pm$ 0.022	1.907 $\pm$ 0.027	1.883 $\pm$ 0.032

Table 10: Results when skills graph for Natural Instructions learned on 125M parameter model is used for data selection with a 1.3B model. We see that SKILL-IT on average still outperforms random and skill-stratified sampling, even though the edges used by SKILL-IT are not derived from the larger model.

no skill influences another skill. We refer this approach as “No graph”. Note that the opposite case of a complete graph recovers skill-stratified sampling, which we already have as a baseline.

Second, instead of sampling over multiple rounds and weighting according to the loss of each skill, we study the effect of setting $T = 1$, which only uses a softmax on A to yield static weights on the skills. We refer to this approach as “Static”. We omit results on Natural Instructions continual pre-training, since SKILL-IT uses $T = 1$ and using no graph with $T = 1$ recovers skill-stratified sampling. Intuitively, we expect the static version of SKILL-IT to perform somewhat well unless there is significant discrepancy among the losses (e.g. in synthetics where the loss on one skill can be close to 0 while the other is not, versus in Natural Instructions where all losses decrease consistently). For both ablations, we sweep over values of $\eta = [0.1, 0.2, 0.5, 0.8]$.

Figure 26 shows the comparison between SKILL-IT and no graph on the continual pre-training LEGO experiment, and Figure 27 shows the comparison between SKILL-IT and a static approach. We see that both the graph and the online dynamics of SKILL-IT are important for its performance. In particular, using no graph results in allocating significant weight to harder skills early on, even though many of them have easier prerequisite skills (such as skill 3 having edges to skills 1 and 2). Using a static graph results in consistent allocation of significant weight to prerequisite skills even after their validation losses converge to near 0, and thus the harder skills that have higher loss are not learned quickly afterwards.

We perform the same ablation on the Addition dataset—the results for this are shown in Figures 28 and Figure 29. We find that these simple baselines, including using a static graph and no graph perform similarly to SKILL-IT on average across all skills—while SKILL-IT performs the best on skill 2 compared to vanilla multiplicative weights, and SKILL-IT performs the best on skill 1 compared to a static graph. This suggests that Addition is somewhat easier than the other datasets that we consider, as SKILL-IT still outperforms other baselines, as shown in Figure 4.

Figure 30 compares SKILL-IT, no graph, and static data selection for the LEGO fine-tuning experiment. No graph can be interpreted as allocating equal weight to all training skills not equal to the target skill, and varying this weight versus the weight on the target skill. While SKILL-IT and setting $T = 1$ behave similarly, we see that SKILL-IT is slightly better than using no graph. For instance, SKILL-IT obtains a validation loss of 0.05 in 2000 steps, compared to 2050-2200 steps when using no graph.

Figure 31 and 32 compare SKILL-IT, no graph, and static data selection for the Natural Instructions fine-tuning experiments. For both Spanish QG and stance detection, SKILL-IT attains lower loss than using no graph or using $T = 1$ round.

Figure 33 compares SKILL-IT and static data selection for the Natural Instructions out-of-domain experiment. SKILL-IT